

# Marketing Campaign & ROI Analytics - Final Report

Prepared for Kavali Harshavardhan. Portfolio simulation, not a real client engagement.

## Scope and definitions

2025 campaign-day data, 10 campaigns across 5 channels; currency INR. All data is generated with seed 42. Each acquisition is an attributed first purchase, not necessarily a globally unique customer. Revenue is simulated last-click attribution within a 7-day window; no multi-touch or causal inference is possible. Contribution profit before marketing equals revenue less simulated variable product/fulfilment costs (55% margin). Spend is total simulated campaign marketing cost. Net contribution is contribution profit minus spend; it excludes corporate overhead, tax and fixed costs.

## Data quality

3,670 raw rows; 20 exact duplicates removed; 15 invalid rows quarantined; 3,635 analyzed rows. Channel labels normalized. Missing spend is excluded rather than assumed to be zero. Exclusions affect totals and could bias rankings in real data.

## Results

Total spend: INR 17,946,461.73. Attributed revenue: INR 157,487,468.50. Net contribution: INR 68,671,646.55. Overall ROAS: 8.78x. Profit-based ROI: 382.6%. CPA: INR 222.84.

Highest observed ROI: Email (2104.5%); lowest: Display (21.1%). These patterns partly reflect the generator's channel assumptions, so they demonstrate analysis methods rather than real channel superiority.

## Recommendations

1. Test a small, capped budget increase for Email; track marginal CPA and contribution, audience saturation and unsubscribe rates.
2. Audit Display targeting, creatives and landing pages before reducing budget; check assisted conversions and brand objectives.
3. Run a randomized holdout or geographic experiment before claiming incremental lift. Last-click ROAS cannot establish causality.
4. Reconcile attributed revenue with orders and refunds; replace assumed margin with actual product-level costs before financial decisions.

## Validation

SQL and Python channel spend, revenue and ROI reconcile to numerical tolerance. Daily records pass nonnegative values, funnel ordering and unique business-key checks. Overall ratios use sums, never average campaign ratios. Undefined ratios are null, not zero.

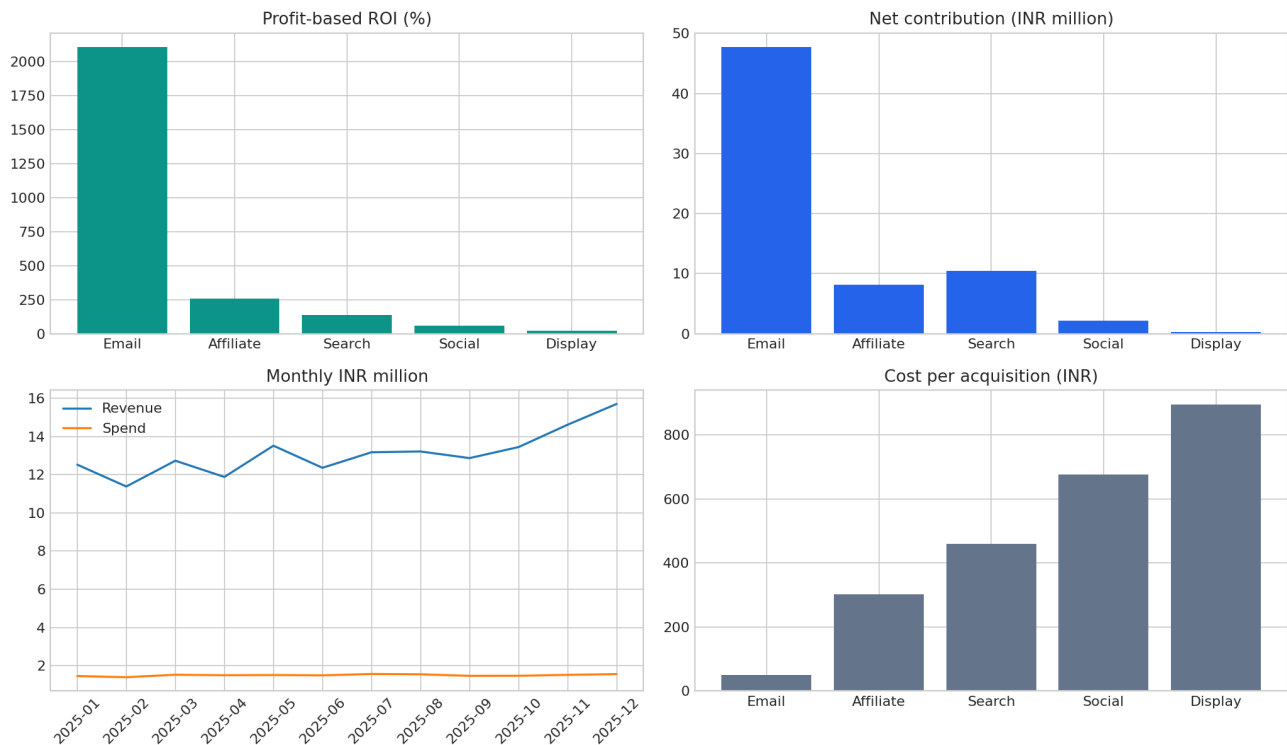
## Limitations

No customer-level retention, CLV or cross-channel deduplication. No actual incrementality, seasonality proof or optimization guarantee. Retention is an audience label only. Quarantined data must be investigated before production use.

# Dashboard and project workflow

Synthetic data | Reporting period: 2025 | Currency: INR

## Marketing performance | SYNTHETIC DATA | 2025 | INR



## End-to-end workflow

Raw data > Cleaning > SQLite and SQL > Python EDA > KPI calculation > Campaign comparison > Excel and dashboard > Insights > Recommendations > Report > GitHub and resume

## Deliverables

The accompanying ZIP includes raw/clean/quarantined CSVs, SQL queries and results, SQLite database, reproducible Python script, EDA notebook, Excel scorecard, interactive offline HTML dashboard, Power BI setup assets, README and interview notes.

## KPI definitions

CTR = clicks / impressions; CPC = spend / clicks; conversion rate = acquisitions / clicks; CPA = spend / acquisitions; ROAS = revenue / spend; ROI = (contribution profit before marketing - spend) / spend. Totals use ratios of sums. Zero denominators return null.

# Complete workflow explained

## 1. Define the business problem

Compare which campaigns produce the highest contribution after marketing cost. Ask which channels have efficient acquisition costs, which campaigns underperform, and what budget experiments should be tested. This project is descriptive analytics, not a causal budget optimizer.

## 2. Collect and understand the data

The generator creates daily records for 10 campaigns across Search, Social, Email, Display and Affiliate. Each record contains date, campaign, audience, spend, impressions, clicks, acquisitions, revenue and contribution profit. The reporting period is 2025 and currency is INR. Data is synthetic and reproducible with seed 42.

## 3. Clean the raw data

Read data/raw\_campaigns.csv. Remove 20 exact duplicates, standardize channel labels, parse dates and numeric fields, and quarantine 15 records with missing spend or invalid funnel values. Check clicks do not exceed impressions and acquisitions do not exceed clicks. Save valid records to data/clean\_campaigns.csv and rejected records to data/quarantined\_rows.csv. Do not replace missing spend with zero.

## 4. Load SQLite and analyze SQL

The Python pipeline loads the clean data into outputs/marketing.sqlite as campaign\_daily. sql/analysis.sql groups by channel, ranks campaign ROI using DENSE\_RANK, calculates month-over-month revenue with LAG, and compares audiences. The query outputs are saved as CSVs for inspection.

## 5. Explore data in Python

Pandas summarizes spend, clicks, acquisitions, revenue and contribution by channel, campaign and month. The EDA notebook shows descriptive statistics, missing-value checks and monthly changes. Matplotlib creates ROI, contribution, revenue/spend trends and CPA charts. Explore patterns before making recommendations; do not interpret generated correlations as real-market evidence.

## 6. Calculate the KPIs

Compute metrics from summed numerators and denominators. CTR measures click engagement, CPC measures click cost, conversion rate measures click-to-acquisition performance, CPA measures acquisition cost, ROAS measures revenue per marketing rupee, and ROI measures net contribution per marketing rupee. ROAS and ROI answer different questions.

## 7. Compare campaigns and channels

Use channel\_summary.csv and campaign\_summary.csv to compare scale, efficiency and contribution together. A small campaign with high ROI may not have capacity for a large budget increase. Review the audience mix and month filters. Email ranks highest in this simulation because its generator assumptions include low CPC and strong conversion.

## 8. Build Excel and dashboards

The Excel workbook contains clean source records and a formula-based channel scorecard using SUMIF and ratios. Editing source values updates the scorecard and linked chart. The offline HTML dashboard provides month/channel filters. Power BI assets supply CSV import, measures and a theme; create the native report in

Power BI Desktop using the guide.

## 9. Turn findings into actions

Propose a capped Email budget test and audit Display creatives, targeting and landing pages. Confirm actual costs, refunds and attribution before operational decisions. Use a randomized holdout or geographic experiment to measure incremental lift. These are proposed actions, not executed improvements.

## 10. Validate and document

Python and SQL channel spend, revenue and ROI are reconciled to numerical tolerance. Check unique date/campaign keys, nonnegative values and valid funnel ordering. Keep the cleaning log, data dictionary and report so reviewers can trace assumptions and exclusions.

## 11. Publish and explain

Upload the extracted project files to GitHub, confirm the README and dashboard image render, and include the repository URL on your resume. Describe it as a synthetic portfolio project. Review the code before claiming independent proficiency and do not claim actual savings or a completed Power BI report until that work is done.

# Run the project and understand the files

## Run in VS Code

Extract the ZIP, open the marketing-roi folder in VS Code, and open a terminal in that folder. Create an optional virtual environment with `python -m venv .venv`. On Windows, activate it with `.venv\Scripts\activate`. Run `python -m pip install -r requirements.txt`, then `python src/analyze.py`. Open notebooks/EDA.ipynb using a Python/Jupyter environment for the exploratory analysis.

## What the pipeline regenerates

`src/analyze.py` reads the existing raw dataset and recreates clean/quarantine CSVs, SQLite database, SQL result CSVs, aggregate KPI CSVs, `dashboard.png`, `dashboard.html` and `FINAL_REPORT.md`. If `raw_campaigns.csv` is absent, it generates the seeded synthetic dataset. It does not automatically refresh Excel or create a native Power BI report.

## Quick demonstration

First show `outputs/dashboard.html` and select a channel or month. Explain why KPI cards change using filtered sums. Then open the Excel workbook and trace channel totals back to Clean data. Show one SQL query, the Python reconciliation checks and the final recommendations. Finish by explaining limitations and the proposed holdout experiment.

## Numerical example

For an illustrative campaign with INR 10,000 spend, INR 30,000 revenue and INR 16,500 contribution before marketing,  $ROAS = 30,000 / 10,000 = 3.0x$ . Net contribution =  $16,500 - 10,000 = \text{INR } 6,500$ .  $ROI = 6,500 / 10,000 = 65\%$ . With 50 acquisitions,  $CPA = \text{INR } 200$ . This example is separate from the dataset totals.

## What is complete and what remains

Completed: sample data, cleaning, SQL/Python analysis, executed EDA summaries, Excel scorecard, charts, offline interactive dashboard, report and documentation. Remaining: upload to your GitHub repository and create/save the .pbix in Power BI Desktop. No live marketing systems, automated campaign spending or production deployment are included.

# Publish to GitHub

Account detected: harshakavali81-collab. No new repository was created or pushed from this session because the available GitHub tools do not create repositories.

## Browser upload (no terminal required)

1. Sign in to GitHub and create a repository named `marketing-campaign-roi-analytics`.
2. Choose Public if you want recruiters to view it. Do not initialize a README if using the terminal method below.
3. Extract the project ZIP. Open the `marketing-roi` folder.
4. On the repository page, choose Add file > Upload files. Drag the folder's contents, preserving subfolders, and commit. Do not upload only the ZIP.
5. Confirm the README renders and the dashboard image appears.

## Git terminal method

After creating an empty repository, run from the extracted project folder:

```
...
```

```
git init -b main
```

```
git add .
```

```
git commit -m "Add marketing ROI analytics portfolio project"
```

```
git remote add origin https://github.com/harshakavali81-collab/marketing-campaign-roi-analytics.git
```

```
git push -u origin main
```

```
...
```

Use your normal GitHub sign-in flow when prompted. Never put access tokens into source files. If you already initialized a repository or set a remote, inspect `git status` and `git remote -v` first rather than reinitializing blindly.

Description: End-to-end synthetic marketing campaign analytics with Python, SQLite, Excel, an interactive dashboard and Power BI setup assets.

Suggested topics: data-analysis, marketing-analytics, sql, python, power-bi, excel.

# Resume and interview notes

Use these only after reviewing and understanding the work. This is a portfolio simulation, not employment experience.

## Resume bullets

- Analyzed 3,635 synthetic campaign-day records using Python and SQL, calculating CTR, CPA, ROAS and profit-based ROI across five marketing channels.
- Built a formula-based Excel scorecard and interactive dashboard; reconciled SQL/Python totals and documented data-quality exclusions and budget-testing recommendations.

After completing the native Power BI report, you may add: Created a Power BI report with DAX measures and channel/month filters to compare marketing efficiency.

## 60-second explanation

I worked on a simulated marketing analytics project to compare campaign profitability. I cleaned duplicated and invalid records, analyzed the clean data in SQLite and Python, and calculated metrics from aggregated totals. I separated ROAS, which measures revenue against spend, from ROI, which uses contribution profit. I built an Excel scorecard and browser dashboard. Email had the highest ROI in the generated dataset, but that reflects the simulation assumptions. I recommended a capped experiment and attribution checks before making real budget changes.

## Questions to prepare

1. Why quarantine missing spend? Zero imputation would overstate returns.
2. Why not average campaign ROI? It weights small campaigns the same as large ones; ratio of total profit to total spend gives the portfolio ROI.
3. Why can high ROAS still be unprofitable? Product and fulfillment costs can absorb revenue.
4. Does attribution prove lift? No; incremental impact needs a control group or experiment.
5. What would improve the project? Actual order/refund data, product margins, customer deduplication, attribution comparison and experiments.
6. How was accuracy checked? SQL/Python reconciliation, unique date/campaign keys and nonnegative funnel consistency checks.

# Build the Power BI report

Native .pbix creation requires Power BI Desktop. These assets are supplied for that final step and have not been executed in Power BI here.

1. Import data/clean\_campaigns.csv using Text/CSV. Name the table Campaigns. Alternatively use import.pq in a blank query and set its path.
2. Check date is Date; spend/revenue/contribution\_profit are Decimal Number; impression/click/acquisition counts are Whole Number; other fields are Text. The single fact table is at date/campaign grain; no relationships are required for this report.
3. Create each measure in measures.dax separately. Format currency measures as INR, rate measures as percentages, and ROAS as 0.00"x".
4. Import theme.json via the report theme controls.
5. Create Overview page: cards for spend, revenue, net contribution, CPA, ROAS, ROI; line chart with month on X and revenue/spend on Y; clustered bar chart with channel and ROI; slicers for month, channel, audience.
6. Create Campaigns page: table with campaign\_id, spend, acquisitions, CPA, ROAS, ROI; bar chart with channel and net contribution; scatter chart with CPA on X, ROI on Y, spend as size and campaign\_id as detail.
7. Create Findings page: add text from docs/FINAL\_REPORT.md, including synthetic-data disclosure, assumptions and proposed holdout test.
8. Add visible titles with units and a synthetic-data banner on every page. Ensure slicers interact with all intended visuals.
9. With all filters cleared, reconcile cards to outputs/overall.csv: spend 17,946,461.73; revenue 157,487,468.50; ROI 382.6473%; ROAS 8.7754x.
10. Filter Email and reconcile with outputs/channel\_summary.csv. Save as Marketing\_ROI\_Analytics.pbix and add it to your repository if desired.

No customer table exists. Do not use DISTINCTCOUNT(acquisitions) or label acquisition totals as unique customers. Blank ratios should remain blank. The month field is YYYY-MM and sorts chronologically.